Pocket Link



File: link.kicad_sch

POCKET LINK rev A – USB-C phone adapter for the Pocket voice agent

Plugs into the phone. Phone sees a USB headset (call audio) + a data interface (Pocket app).

The app bridges call audio <-> voice agent WebSocket using the phone internet. No radio, no battery.

RP2040 + 2 MB QSPI flash + AP2112K 3.3 V LDO, powered from phone VBUS.

4-layer 0.8 mm (plug straddles the board edge). Fab: JLCPCB, see fab/ folder.

Pocket		
Sheet: /		
File: pocket-link.kicad_sch		
Title: Pocket Hub-8		
Size: A4	Date: 2026-10-06	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/2

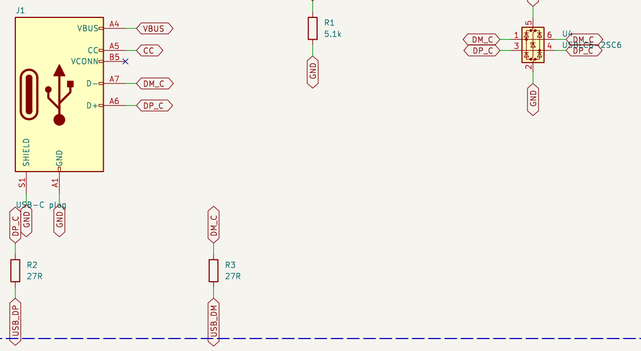
Phone = USB host. J1 plugs into the phone; CC has Rd (5.1k) so the phone supplies VBUS and enumerates us as a device.

RP2040 firmware (TinyUSB) = composite device: UAC headset (16 kHz mono in/out) + vendor bulk data interface for the Pocket app.

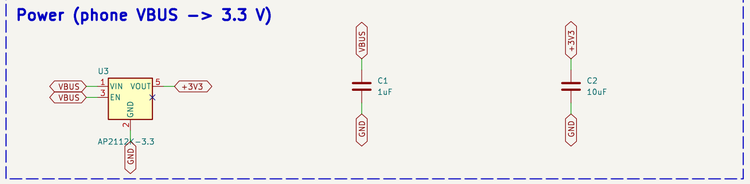
Blank flash -> RP2040 USB bootloader appears as a drive: drag the .uf2. To force bootloader later: short TP BOOT to GND while plugging in.

RP2040 support circuit follows the Raspberry Pi RP2040 hardware design guide (1k on XOUT, 27R USB series, 1uF on VREG).

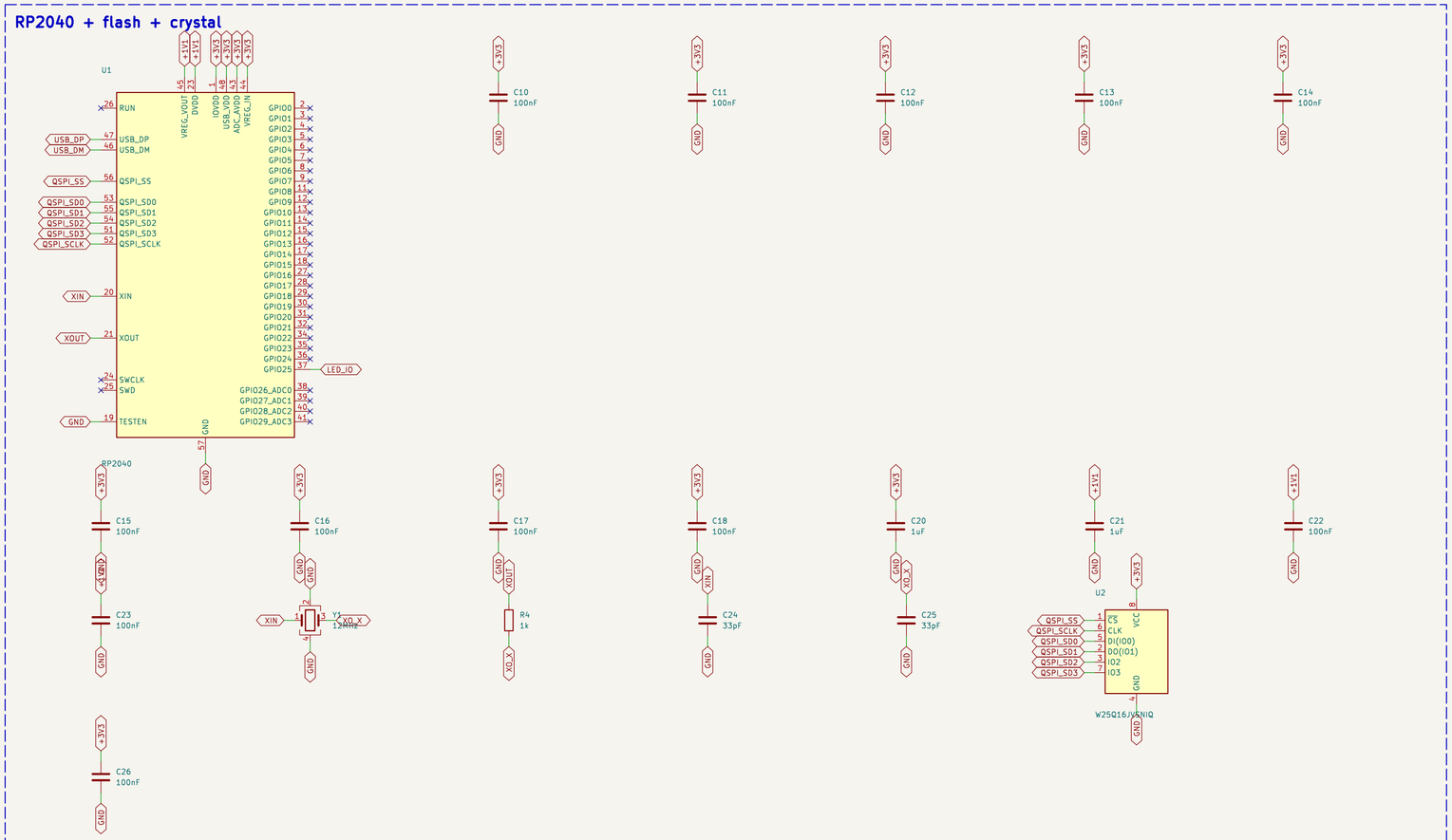
USB-C plug + ESD



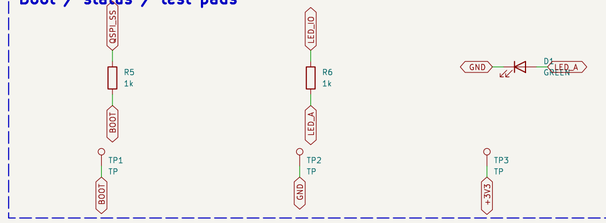
Power (phone VBUS -> 3.3 V)



RP2040 + flash + crystal



Boot / status / test pads



Generated from design.py — edit the source, not this file

USB-C phone adapter: USB headset + data channel

Pocket

Sheet: /Pocket Link/

File: link.kicad_sch

Title: Pocket Link — Pocket Link — USB-C phone adapter

Size: A2	Date: 2026-10-06	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/2